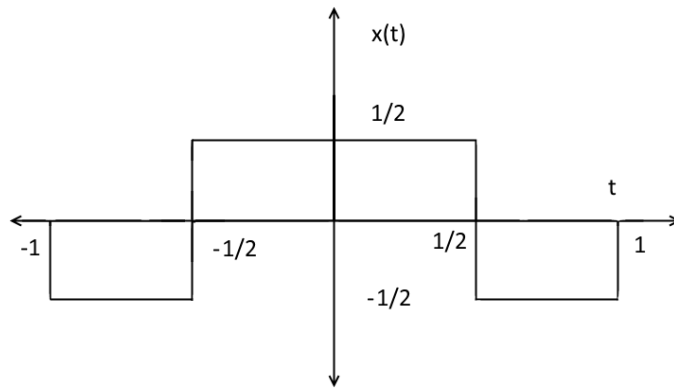


Problem sheet–4

1. Fourier series representation of a signal

Determine a_k for the following signal $x(t)$ with period 2.



2. Fourier series representation of a chess board

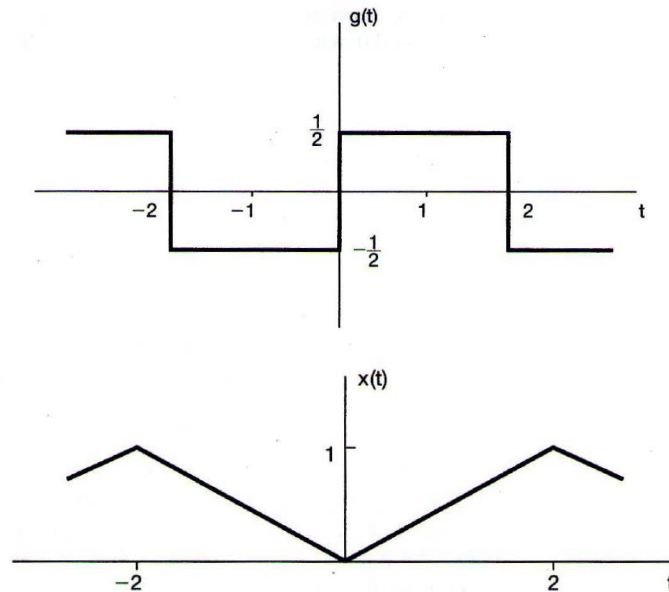
A chess board image has to be represented using a Fourier series. Think about a possible representation.
(Hint: image is a two dimensional signal)

3. Using properties of Fourier Series

Consider the signals $g(t)$ and $x(t)$ given below of fundamental period $T=4$.

Note that the derivative of signal $x(t)$ is same as signal $g(t)$.

Find the Fourier coefficients of signal $g(t)$. Using the derivative property of Fourier Series coefficients, find the same for signal $x(t)$.



4. Fourier representation of an Impulse Train

- a) Find the Fourier series representation of the impulse train given by

$$x(t) = \sum_{k=-\infty}^{\infty} \delta(t - kT)$$

- b) Represent the derivative of $g(t)$ in the question above in terms of a shifted impulse train and hence find the Fourier coefficients of $g(t)$ in terms of Fourier Coefficients of $x(t)$. Verify it with your answer for question 3.

5. Relationship between Fourier coefficients

Let $x_1(t)$ be a continuous-time periodic signal with fundamental frequency ω_1 and Fourier coefficients a_k . Given that

$$x_2(t) = x_1(1 - t) + x_1(t - 1)$$

how is the fundamental frequency ω_2 of $x_2(t)$ related to ω_1 ? Also, find a relationship between the Fourier series coefficients b_k of $x_2(t)$ and the coefficients a_k . You may use the properties of the Fourier series coefficients.

6. Signal Determination from Fourier coefficient properties

Suppose We are given the following information about a signal $x(t)$:

- $x(t)$ is real and odd.
- $x(t)$ is periodic with period $T = 2$ and has Fourier coefficients a_k .
- $a_k = 0$ for $|k| > 1$.
- $\frac{1}{2} \int_0^2 |x(t)|^2 dt = 1$

Specify 2 different signals with the above properties.

7. Filter output evaluation using Fourier Series

Consider a continuous-time LTI system S whose frequency response is

$$H(\omega) = \begin{cases} 1, & |\omega| \geq 250 \\ 0, & \text{otherwise} \end{cases}$$

When the input to this system is a signal $x(t)$ with fundamental period $T = \pi/7$ and Fourier series coefficients a_k , it is found that the output $y(t)$ is identical to $x(t)$. For what values of k is it guaranteed that $a_k = 0$?

8. Properties of Fourier series coefficient

Let $x(t)$ be a periodic signal with fundamental period T and Fourier series coefficients a_k . Derive the Fourier series coefficients of each of the following signals in terms of a_k :

- (a) $x(t - t_0) + x(t + t_0)$
- (b) $\text{Ev}(x(t))$
- (c) $\text{Re}\{x(t)\}$
- (d) $\frac{d^2 x(t)}{dt^2}$
- (e) $x(3t - 1)$ [for this part, first determine the period of $x(3t - 1)$]

9. Signal Determination from Fourier coefficient properties

Suppose we are given the following information about a continuous-time periodic signal with period 3 and Fourier coefficients a_k :

- a) $a_k = a_{k+2}$
- b) $a_k = a_{-k}$
- c) $\int_{-0.5}^{0.5} x(t) dt = 1$
- d) $\int_1^2 x(t) dt = 2$

Determine $x(t)$

10. Even and Odd Harmonic Signals

- a) A continuous-time periodic signal $x(t)$ with period T is said to be odd harmonic if, in its Fourier series representation

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk(2\pi/T)t}$$

$a_k = 0$ for every even integer k .

- b) Show that if $x(t)$ is odd harmonic, then

$$x(t) = -x\left(t + \frac{T}{2}\right)$$

- c) Show that if $x(t)$ satisfies eq. above, then it is odd harmonic.
- d) Suppose that $x(t)$ is an odd-harmonic periodic signal with period 2 such that

$$x(t) = t \quad \text{for } 0 < t < 1.$$

Sketch $x(t)$.

- e) Analogously, to an odd-harmonic signal, we could define an even-harmonic signal as a signal for which $a_k = 0$ for k odd in the Fourier series representation. Could T be the fundamental period for such a signal? Explain your answer.
- f) More generally, show that T is the fundamental period of $x(t)$ in eq. in part(a) if one of two things happens:
 - i. Either a_1 or a_{-1} is nonzero; or
 - ii. There are two integers k and l that have no common factors and are such that both a_k and a_l are nonzero.

11. Signal Determination from Fourier coefficient properties

Suppose we are given the following information about a signal $x(t)$:

- a) $x(t)$ is a real signal.
- b) $x(t)$ is periodic with period $T = 6$ and has Fourier coefficients a_k .
- c) $a_k = 0$ for $k = 0$ and $k > 2$.

d) $x(t) = -x(t - 3)$.

e) $\frac{1}{6} \int_{-3}^3 x(t) dt = \frac{1}{2}$

f) a_1 is a positive real number.

Show that $x(t) = A \cos(Bt + C)$, and determine the values of the constants A, B, and C.

12. Orthogonal representation of signals

Two functions $u(t)$ and $v(t)$ are said to be orthogonal over the interval (a,b)

$$\int_a^b u(t)v^*(t)dt = 0$$

If, in addition,

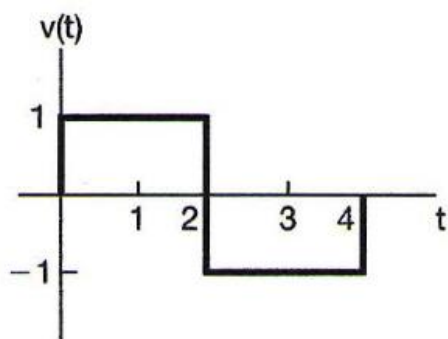
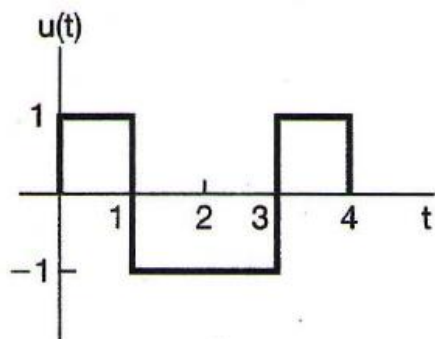
$$\int_a^b |u(t)|^2 dt = 1, \quad \int_a^b |v(t)|^2 dt = 1$$

the functions are said to be normalized and hence are called orthonormal. A set of functions $\{\varphi_k(t)\}$ is called an orthogonal (orthonormal) set if each pair of functions in the set is orthogonal (orthonormal).

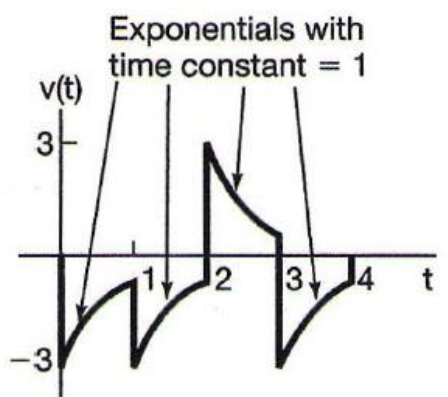
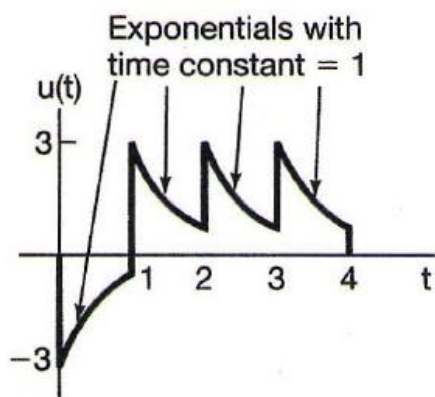
- Consider the pairs of signals $u(t)$ and $v(t)$ depicted in Figure below. Determine whether each pair is orthogonal over the interval (0, 4).
- Are the functions $\sin(mw_0t)$ and $\sin(nw_0t)$ orthogonal over the interval (0, T), where $T = 2\pi/w_0$? Are they also orthonormal?
- Repeat part (b) for the functions $\varphi_m(t)$ and $\varphi_n(t)$, where $\varphi_k(t) = \frac{1}{\sqrt{T}}[\cos(kw_0t) + \sin(kw_0t)]$.
- Show that the functions $\varphi_k(t) = e^{jkw_0t}$ are orthogonal over any interval of length $T = 2\pi/w_0$. Are they orthonormal?
- Let $x(t)$ be an arbitrary signal, and let $x_o(t)$ and $x_e(t)$ be, respectively, the odd and even parts of $x(t)$. Show that $x_o(t)$ and $x_e(t)$ are orthogonal over the interval $(-T, T)$ for any T.
- Show that if $\{\varphi_k(t)\}$ is a set of orthogonal signals over the interval (a, b), then the set $\{\frac{1}{\sqrt{A_k}}\varphi_k(t)\}$, where

$$A_k = \int_a^b |\varphi_k(t)|^2 dt$$

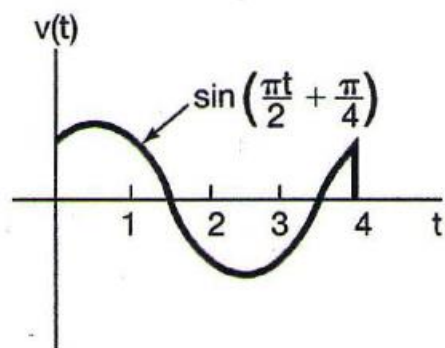
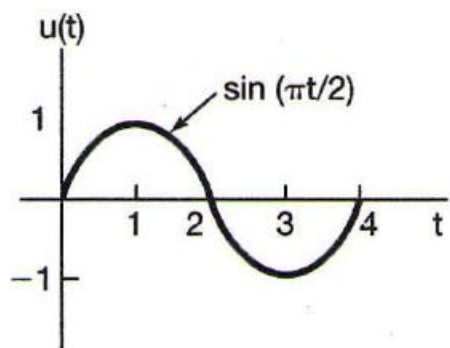
is orthonormal.



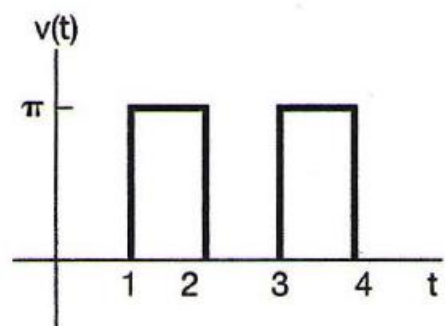
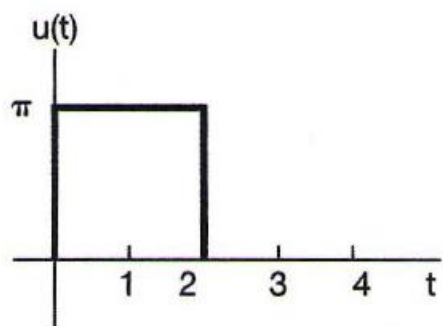
(a)



(b)



(c)



(d)